

ASSIGNMENT-- 1.3

Q1. Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

Feature	DDR3	DDR4	DDR5
1. Release Year	2007	2014	2020
2. Speed	800–2133 MT/s	1600–3200 MT/s	4800–8400+ MT/s
3. Voltage	1.5V	1.2V	1.1V
4. Power Use	High	Medium	Low (Most Efficient)
5. Performance	Good	Better	Best
6. Data Transfer	Slow	Fast	Very Fast
7. Maximum Capacity	Up to 16 GB per module	Up to 64 GB per module	Up to 128 GB
8. Gaming	Good for old games	Good for modern games	Best for latest games
9. Multitasking	Basic	Smooth	Very Smooth
10. Compatible CPUs	Older CPUs	Newer CPUs	Latest CPUs
11. Price	Cheapest	Medium	Most Expensive
12. Typical Use	Old PCs, office work	Home, office, gaming	High-end gaming, AI, video editing, workstations

Q2. Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.

 Hint: Refer to your device manual or a trusted YouTube guide if unsure about RAM removal and installation steps.



➤ **Simple Steps to Remove and Reinstall RAM:**

1. **Turn off** the computer or laptop.
2. **Unplug the power cable** and charger.
3. **Open the back cover** (laptop) or side panel (desktop).
4. **Find the RAM slot** on the motherboard.
5. **Push the two side clips** outward. The RAM will pop up.
6. **Remove the RAM** carefully by holding the edges.
7. **Take a photo** of the RAM slot and RAM module.
8. **Insert the RAM back** into the same slot.
9. **Press gently** until the clips lock automatically.
10. **Close the cover** and reconnect the power.
11. **Turn on the computer.**
12. **Check that the computer starts normally.** If it boots successfully, the RAM is installed correctly.

Q3. List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

➤ **Three Key Differences**

- **Speed:** **HDDs** are slow. **SSDs** are fast. **NVMe** drives are super-fast.
- **Parts:** **HDDs** have moving parts inside like a spinning disk. **SSDs** and **NVMe** have no moving parts and use microchips to save data.
- **Connection:** **HDDs** and **SSDs** use old, slower **SATA wires**. **NVMe** plugs directly into the fastest slot on the motherboard (**PCIe**).

➤ **Best Daily Use Case for Each**

1. HDD (Hard Disk Drive)

- **Best Use: Saving big files (Photos, Movies, Backups)**
- **Why:** It gives you a lot of storage space for very little money. You do not need high speed just to store old files.

2. SSD (SATA Solid State Drive)

- **Best Use: Daily apps (Google Chrome, Word, Excel)**
- **Why:** It makes your computer turn on fast. Your daily apps will open instantly without slowing down your computer.

3. NVMe (Fastest Storage)

- **Best Use: Heavy work (Video Editing or Heavy Gaming)**
- **Why:** Moving giant files or loading massive video games takes a lot of power. NVMe moves gigabytes of data in just one second.

**Q4. Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly.

Constraint: Only perform this task if you have access to a device you are allowed to open. If not, find and share a YouTube video link showing the process for your laptop/PC model. **

